Dominant Alleles

17A

When a gene for a characteristic can appear in two different forms, these are called alleles.

The basic unit of inheritance is the *gene*. When genes are available in alternative forms, these alternative forms are called *alleles*. The genes on paired chromosomes control the same characteristic and can give identical or conflicting instructions to the cells. If the instructions are different, only one gene will be obeyed. The instructions from one gene will override the other.

The gene that 'wins' is called the *dominant* gene and the one that 'loses' is called the *recessive* gene. Dominant characteristics will always appear in the first generation offspring (F_1). When a dominant trait (allele) is present, it will always mask the presence of a recessive characteristic (allele). The dominant allele is represented by an upper-case (capital) letter and the recessive allele by a lower-case letter. Recessive characteristics only become visible in their 'pure' (*homozygous*) form, when a double dose of that recessive gene is present. Since we receive one gene from each parent, the only way for the recessive to be visible in the *phenotype* (physical characteristic) is if it is the only type of gene passed on by the parents.

In Mendel's experiments he crossed peas producing green seeds with peas producing yellow seeds. 'G' represents an allele for yellow seeds and 'g' represents an allele for green seeds. Study the Punnett square on the right, which shows a cross between two heterozygous parent plants. (See worksheets 21 and 22 about Punnett squares.) It is called a monohybrid cross because it only involves one characteristic.

	G	g
G		
g		

1 Complete the Punnett square.

2 What will be the ratios of the phenotypes of the F_1 offspring?

_____

3 Give the genotypes of a homozygous green plant. _____

4 Give the genotypes of a homozygous yellow plant. _____

Partial dominance

Sometimes a cross does not result in clear-cut differences between parents and offspring. Characteristics sometimes appear to blend in the F_1 generation. The 'Four O'Clock Plant' can occur in three colours: red, pink and white. The alleles present are R and r. A red plant will be RR, a white plant will be rr and a pink plant will be Rr. Neither characteristic is dominant.

5 Complete the Punnett square on the right, showing the results of a cross between a red plant and a white plant.

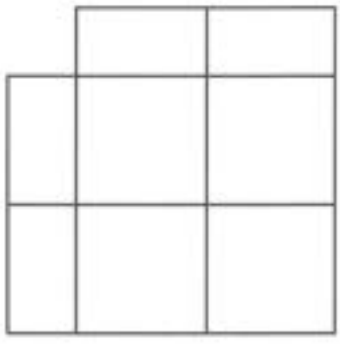
6 What will be the ratios of the phenotypes of the F_1 offspring?

_____

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7 Complete the Punnett square on the right, showing the results of a cross between two pink plants.



8 What will be the ratios of the phenotypes of the offspring of two pink plants?

_____

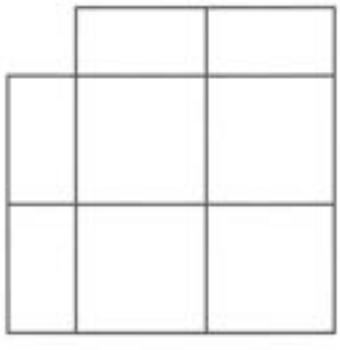
9 Is the 'blending' occurring in the genotypes or in the phenotypes? Explain your answer.

_____

_____

Codominance

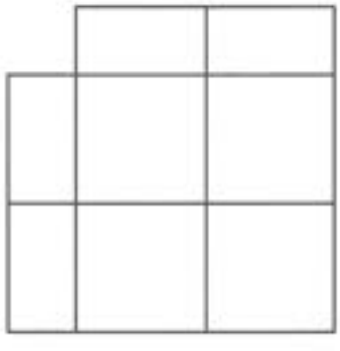
Short-horn cattle can be (among other colours) red-haired or white-haired. If a red-haired animal is crossed with a white-haired animal, the resulting offspring are not completely red, white or pink. They have *both* red and white hairs and are said to be *roan* coloured. The alleles for red and white both produce an effect when they are present together. We represent codominant alleles as both having capital letters, for example R_1 = red, and R_2 = white.



10 Complete the Punnett square on the right, showing the results of a cross between two roan cattle.

11 What will be the ratios of the phenotypes of the offspring of two roan cattle?

_____



12 Complete the Punnett square on the right, showing the results of a cross between a roan cow and a white bull.

13 What will be the ratios of the phenotypes of the offspring of a roan cow and a white bull?

_____

_____

One example of codominance in humans is the inheritance of some blood groups.

14 How many characteristics are involved in a 'dihybrid cross'? _____

15 What organism was used by Gregor Mendel in his studies of inheritance? _____

16 What is another name for a 'pure-breeding' organism? _____

17 What is meant by F_1? _____

18 '*Homozygous* means having identical alleles for a characteristic.' True or false? _____

19 'A *phenotype* is a physical characteristic.' True or false? _____

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